

Brute Force Attack Investigation — BruteForce.pcap & auth.log

Objective

Investigate brute-force activity against a compromised web server, identify attacker behavior across HTTP, RDP, and SSH protocols, and map it to MITRE ATT&CK techniques.

Evidence Analyzed

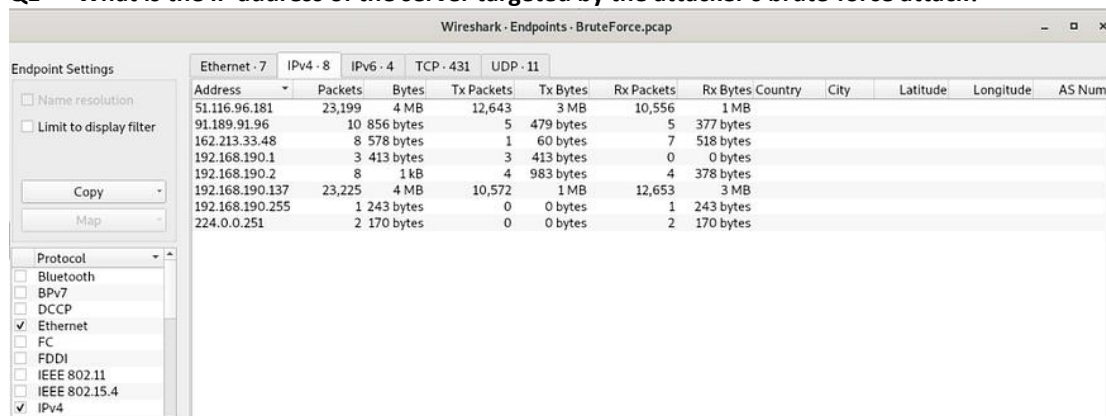
- BruteForce.pcap (packet capture — network traffic)
- auth.log (server authentication log)

Tools & Methods

- Wireshark (packet inspection, protocol filtering, HTTP/RDP analysis)
- Text editor / grep (log parsing, credential search)
- MITRE ATT&CK (adversary behavior classification)

Challenges

Q1 — What is the IP address of the server targeted by the attacker's brute-force attack?

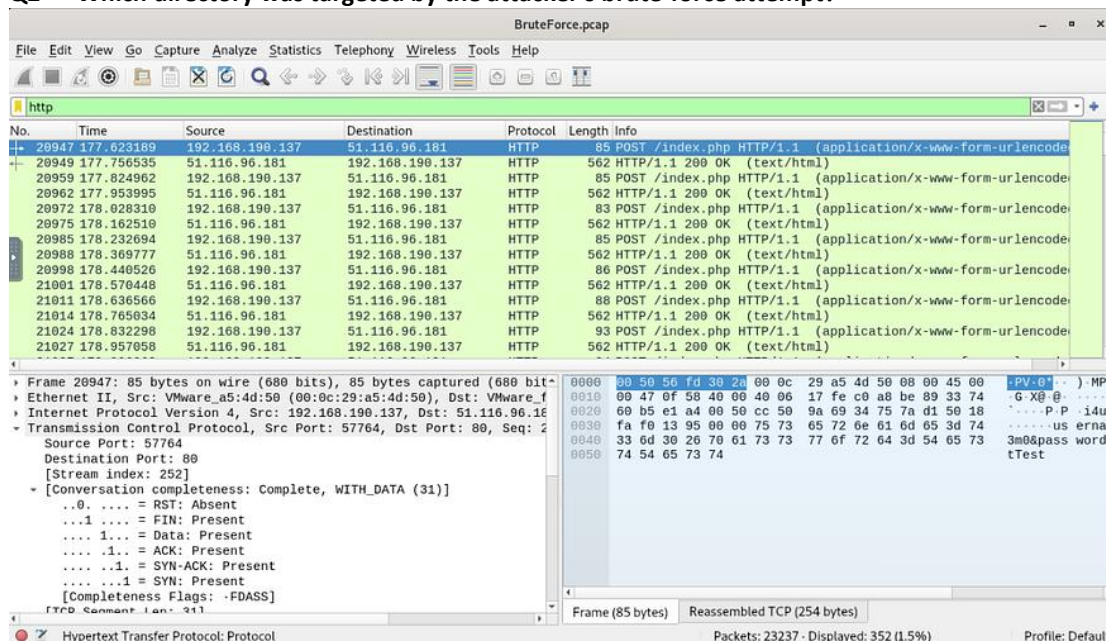


Address	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes	Country	City	Latitude	Longitude	AS Num
51.116.96.181	23,199	4 MB	12,643	3 MB	10,556	1 MB					
91.189.91.96	10	856 bytes	5	479 bytes	5	377 bytes					
162.213.33.48	8	578 bytes	1	60 bytes	7	518 bytes					
192.168.190.1	3	413 bytes	3	413 bytes	0	0 bytes					
192.168.190.2	8	1 kB	4	983 bytes	4	378 bytes					
192.168.190.137	23,225	4 MB	10,572	1 MB	12,653	3 MB					
192.168.190.255	1	243 bytes	0	0 bytes	1	243 bytes					
224.0.0.251	2	170 bytes	0	0 bytes	2	170 bytes					

Using Wireshark Statistics → Endpoints (IPv4), the high-volume target was identified as

Answer: 51.116.96.181

Q2 — Which directory was targeted by the attacker's brute-force attempt?



No.	Time	Source	Destination	Protocol	Length	Info
28947	177.623189	192.168.190.137	51.116.96.181	HTTP	85	POST /index.php HTTP/1.1 (application/x-www-form-urlencoded)
28949	177.756535	51.116.96.181	192.168.190.137	HTTP	562	HTTP/1.1 200 OK (text/html)
28959	177.824962	192.168.190.137	51.116.96.181	HTTP	85	POST /index.php HTTP/1.1 (application/x-www-form-urlencoded)
28962	177.953995	51.116.96.181	192.168.190.137	HTTP	562	HTTP/1.1 200 OK (text/html)
28972	178.028310	192.168.190.137	51.116.96.181	HTTP	83	POST /index.php HTTP/1.1 (application/x-www-form-urlencoded)
28975	178.162510	51.116.96.181	192.168.190.137	HTTP	562	HTTP/1.1 200 OK (text/html)
28985	178.232694	192.168.190.137	51.116.96.181	HTTP	85	POST /index.php HTTP/1.1 (application/x-www-form-urlencoded)
28988	178.369777	51.116.96.181	192.168.190.137	HTTP	562	HTTP/1.1 200 OK (text/html)
28998	178.440526	192.168.190.137	51.116.96.181	HTTP	86	POST /index.php HTTP/1.1 (application/x-www-form-urlencoded)
21001	178.570448	51.116.96.181	192.168.190.137	HTTP	562	HTTP/1.1 200 OK (text/html)
21011	178.636566	192.168.190.137	51.116.96.181	HTTP	88	POST /index.php HTTP/1.1 (application/x-www-form-urlencoded)
21014	178.765034	51.116.96.181	192.168.190.137	HTTP	562	HTTP/1.1 200 OK (text/html)
21024	178.832298	192.168.190.137	51.116.96.181	HTTP	93	POST /index.php HTTP/1.1 (application/x-www-form-urlencoded)
21027	178.957058	51.116.96.181	192.168.190.137	HTTP	562	HTTP/1.1 200 OK (text/html)

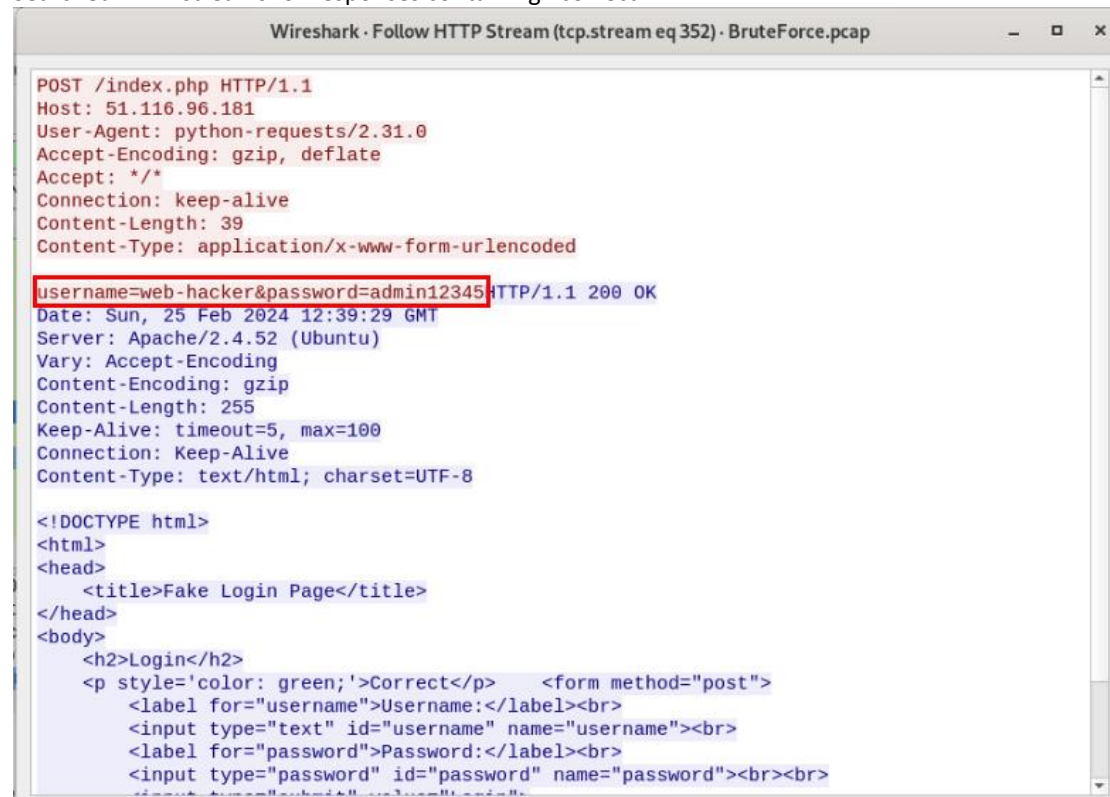
Frame 28947: 85 bytes on wire (680 bits), 85 bytes captured (680 bits) on interface 0
Ethernet II, Src: VMware_a5:4d:50 (00:0c:29:a5:4d:50), Dst: VMware_f
Internet Protocol Version 4, Src: 192.168.190.137, Dst: 51.116.96.181
Transmission Control Protocol, Src Port: 57764, Dst Port: 80, Seq: 2
Source Port: 57764
Destination Port: 80
[Stream index: 252]
[Conversation completeness: Complete, WITH_DATA (31)]
...0... = RST: Absent
...1... = FIN: Present
...1... = Data: Present
...1... = ACK: Present
...1... = SYN-ACK: Present
...1... = SYN: Present
[Completeness Flags: -FDASS]
TCP Segment Len: 81

Filtering for HTTP traffic showed repeated POST requests against

Answer: index.php

Q3 — Identify the correct username and password combination used for login.

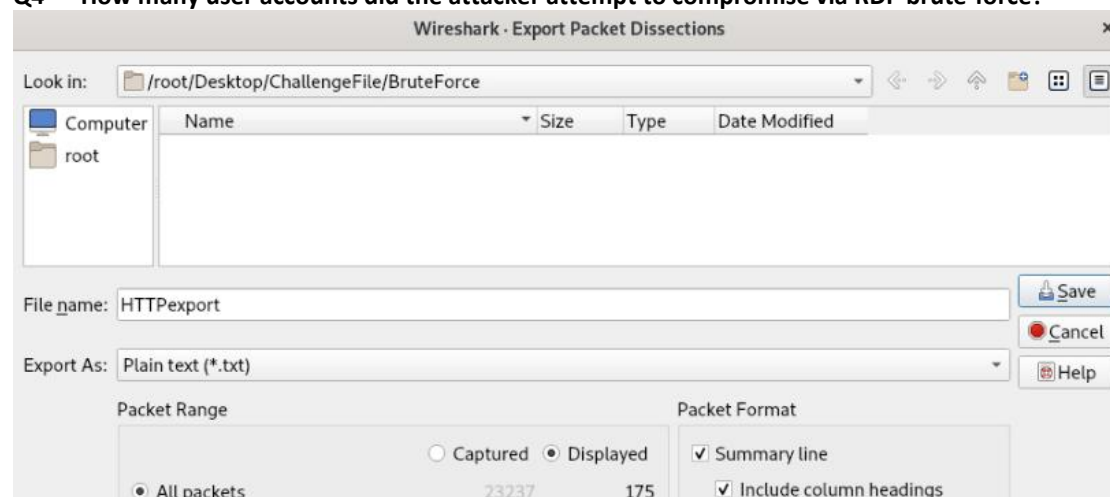
Searched HTTP streams for responses containing >correct.



Found valid POST request revealing:

Answer: Username / Password: web-hacker:admin12345

Q4 — How many user accounts did the attacker attempt to compromise via RDP brute-force?



open the terminal and use grep to search the text file, displaying only the lines matching "username" and then removing any duplicate entries.

cat HTTPexport.txt | grep -i "username" | uniq

```

root@ip-172-31-21-51:~/Desktop/ChallengeFile/BruteForce# cat HTTPexport.txt | grep -i "username" | uniq
Form item: "username" = "t3m0"
Form item: "username" = "MoSalah"
Form item: "username" = "Messi"
Form item: "username" = "web-hacker"
Form item: "username" = "Kareem"
Form item: "username" = "Mostafa"
Form item: "username" = "mmox"

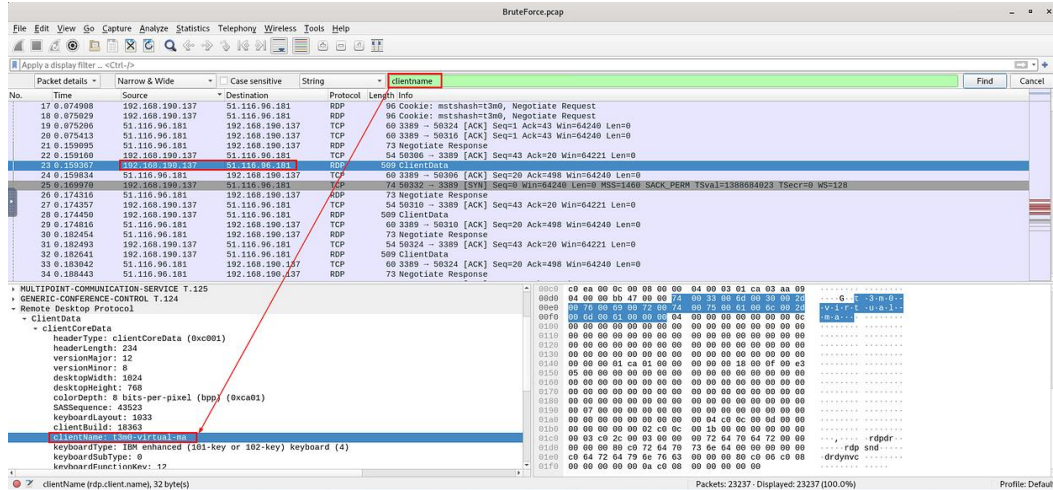
```

Exported POST requests, extracted usernames via grep:

Answer: 7

Q5 — What is the “clientName” of the attacker's machine?

Searched for clientName in Wireshark.



Found in RDP ClientData:

Answer: t3m0-virtual-ma

Q6 — When did the user last successfully log in via SSH, and who was it?

```

21730 Feb 25 11:43:45 chall sudo: root : PWD=/var/lib/waagent/Microsoft.CPLat.Core.LinuxPatchExtension
21731 Feb 25 11:43:45 chall sudo: sbin/waagent --version
21732 Feb 25 11:43:46 chall sudo: pam_unix(sudo:session): session opened for user root(uid=0) by (uid=0)
21733 Feb 25 11:43:46 chall sudo: root : PWD=/var/lib/waagent/Microsoft.CPLat.Core.LinuxPatchExtension-1.6.50 ; USER=root ; COMMAND=/usr/
21734 Feb 25 11:43:46 chall sudo: sbin/waagent --version
21735 Feb 25 11:43:46 chall sudo: pam_unix(sudo:session): session opened for user root(uid=0) by (uid=0)
21736 Feb 25 11:43:46 chall sudo: pam_unix(sudo:session): session closed for user root
21737 Feb 25 11:43:54 chall sshd[981]: Accepted password for mmox from 41.38.160.33 port 54464 ssh2
21738 Feb 25 11:43:54 chall sshd[981]: pam_unix(sshd:session): session opened for user mmox(uid=1000) by (uid=0)
21739 Feb 25 11:43:54 chall systemd-logind[578]: New session 1 of user mmox.
21740 Feb 25 11:44:17 chall sudo: mmox : TTY=pts/0 ; PWD=/home/mmox ; USER=root ; COMMAND=/usr/bin/systemctl status xrdp
21741 Feb 25 11:44:17 chall sudo: pam_unix(sudo:session): session opened for user root(uid=0) by mmox(uid=1000)

```

Parsed auth.log for Accepted password.

Last entry revealed successful login by:

Answer: mmox:11:43:54

Q7 — How many unsuccessful SSH connection attempts were made by the attacker?

```

Open auth.log
1 Feb 24 09:48:50 chall su: pam_unix(su:session): session closed for user root
2 Feb 24 09:48:50 chall sudo: pam_unix(sudo:session): session closed for user root
3 Feb 24 09:49:59 chall sudo: mmox : TTY=pts/0 ; PWD=/var/www/html ; USER=root ; COMMAND=/usr/bin/curl -s https://raw.githubusercontent.com/0x00sec/0x00sec/master/0x00sec.txt
4 Feb 24 09:49:59 chall sudo: pam_unix(sudo:session): session opened for user root(uid=0) by mmox(uid=1000)
5 Feb 24 09:49:59 chall sudo: pam_unix(sudo:session): session closed for user root
6 Feb 24 09:50:09 chall sudo: mmox : TTY=pts/0 ; PWD=/var/www/html ; USER=root ; COMMAND=/usr/bin/nano index.php
7 Feb 24 09:50:09 chall sudo: pam_unix(sudo:session): session opened for user root(uid=0) by mmox(uid=1000)
8 Feb 24 09:52:09 chall sudo: pam_unix(sudo:session): session closed for user root
9 Feb 24 09:52:21 chall sudo: mmox : TTY=pts/0 ; PWD=/var/www/html ; USER=root ; COMMAND=/usr/bin/apt install net-tools
10 Feb 24 09:52:21 chall sudo: pam_unix(sudo:session): session opened for user root(uid=0) by mmox(uid=1000)
11 Feb 24 09:52:26 chall sudo: pam_unix(sudo:session): session closed for user root
12 Feb 24 09:52:43 chall sshd[1759]: Received disconnect from 196.136.60.15 port 13586:11: disconnected by user
13 Feb 24 09:52:43 chall sshd[1759]: Disconnected from user mmox 196.136.60.15 port 13586
14 Feb 24 09:52:43 chall sshd[1657]: pam_unix(sshd:session): session closed for user mmox
15 Feb 24 09:52:43 chall systemd-logind[729]: Session 1 logged out. Waiting for processes to exit.
16 Feb 24 09:52:43 chall systemd-logind[729]: Removed session 1.
17 Feb 24 09:52:53 chall systemd: pam_unix(systemd-user:session): session closed for user mmox
18 Feb 24 10:11:43 chall systemd-logind[612]: New seat seat0.
19 Feb 24 10:11:43 chall systemd-logind[612]: Watching system buttons on /dev/input/event0 (AT Translated Set 2 keyboard)
20 Feb 24 10:11:43 chall sshd[736]: Server listening on 0.0.0.0 port 22.
21 Feb 24 10:11:43 chall sshd[736]: Server listening on *: port 22.
22 Feb 24 10:13:55 chall sshd[1039]: pam_unix(sshd:auth): authentication failure; logname=uid=0 euid=0 tty=ssh ruser= rhost=196.136.60.15 user=mmox
23 Feb 24 10:13:57 chall sshd[1039]: Failed password for mmox from 196.136.60.15 port 32976 ssh2
24 Feb 24 10:14:03 chall sshd[1039]: Failed password for mmox from 196.136.60.15 port 32976 ssh2
25 Feb 24 10:14:08 chall sshd[1039]: Accepted password for mmox from 196.136.60.15 port 32976 ssh2
26 Feb 24 10:14:08 chall sshd[1039]: pam_unix(sshd:session): session opened for user mmox(uid=1000) by (uid=0)

```

Parsed auth.log for Failed password.

Total failed attempts:

Answer: 7480

Q8 —What technique is used to gain access?

Mapped observed activity (systematic guessing across HTTP, RDP, and SSH) to MITRE ATT&CK:

Answer- T1110 – Brute Force

Conclusion

This investigation revealed that the web server (51.116.96.181) was the victim of a coordinated brute-force attack across multiple services. The attacker:

- Hammered the index.php endpoint with HTTP POST login attempts.
- Attempted RDP brute force, leaking their clientName in RDP negotiation.
- Conducted SSH brute-force attacks with hundreds of failed logins before eventually succeeding with a valid account.
- Impact: Unauthorized access was achieved using stolen credentials.

Recommendation:

- Immediately reset all compromised credentials.
- Enforce strong password policies & rate-limiting.
- Monitor and alert on high-volume failed login attempts across all protocols.
- Consider MFA (multi-factor authentication) to harden remote access.